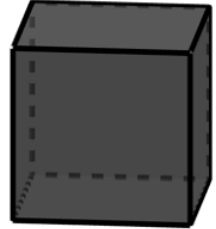


Grade 6 Accelerated
Unit 8
Lesson 1 Homework

Name _____
Date _____
Period _____

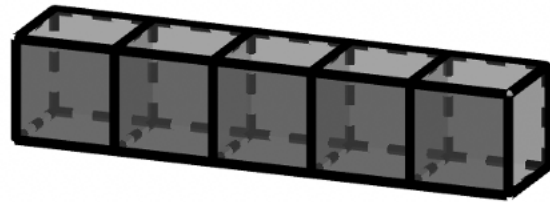
A $\frac{1}{3}$ inch unit cube is shown.

1. Determine the volume of the unit cube. Include the units for the cube's volume.



The diagram shows five unit cubes whose dimensions are $\frac{1}{5}$ inches.

2. The volume of one cube is
_____ cubic inches (in^3).

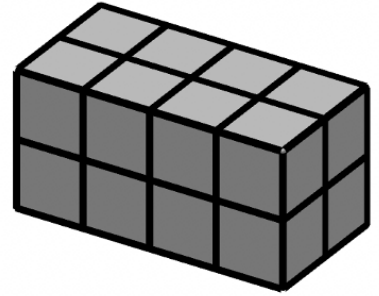


3. The volume of all five cubes is _____ in^3 .

A rectangular prism is shown.

4. How many unit cubes whose dimensions are $\frac{1}{4}$ inch fill up the rectangular prism?

5. Each unit cube whose dimensions are $\frac{1}{4}$ inch has a volume of $\frac{1}{64}$ cubic inches. Find the total volume, in cubic inches, represented in the diagram by multiplying the number of cubes that fill the rectangular prism by the volume of one of the cubes.

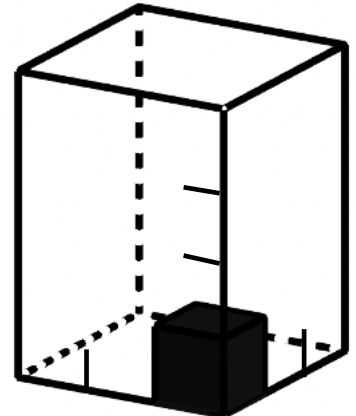


6. Find the total volume represented in the diagram, in cubic inches, using the formula, $V = lwh$.

A rectangular prism with one cube inside is shown.

7. How many cubes will it take to fill the rectangular prism?

8. The rectangular prism has a length of $\frac{12}{5}$ centimeters (cm), a width of $\frac{12}{5}$ cm, and a height of $\frac{16}{5}$ cm as shown. Determine the dimensions of the fractional unit cube.



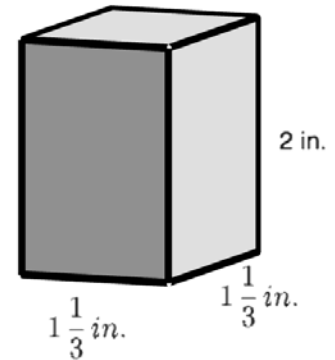
9. Determine the volume of the unit cube.
10. Determine the volume of the rectangular prism.

Grade 6 Accelerated
Unit 8
Lesson 2 Homework

Name _____
Date _____
Period _____

A rectangular prism is shown with units measured in inches.

In questions 1–4, find the volume using cubes with a side length of $\frac{2}{3}$ in.



1. Determine how many cubes will fit along the length, width, and height.

$$1\frac{1}{3} \div \frac{2}{3} = \underline{\hspace{2cm}} \text{ cubes will fit along the length.}$$

$$1\frac{1}{3} \div \frac{2}{3} = \underline{\hspace{2cm}} \text{ cubes will fit along the width.}$$

$$2 \div \frac{2}{3} = \underline{\hspace{2cm}} \text{ cubes will fit along the height.}$$

2. A total of _____ cubes will fit within the rectangular prism.
3. The volume of each cube is $\frac{8}{27} \text{ in}^3$. Multiply the number of cubes needed to fill the prism by the volume of each cube in order to find the volume of the prism. Include units.
4. Instead of using cubes, Kaia uses the formula $V = l \times w \times h$. Fill in the spaces below to complete her work. Include units.

$$V = \underline{\hspace{2cm}} \times \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$$

$$V = \underline{\hspace{2cm}} \text{ in.}^3$$

A baker is packaging her cupcakes into single serve boxes for a special event. Each cube-shaped box has side lengths 3.5 inches (in.). The smaller boxes are then packed into a larger box with dimensions of 10.5 in. $\times$ 7 in. $\times$ 7 in.

5. How many single serve cupcakes can be packed into the larger box?
6. Use the number of cupcakes that can be packed in the larger box to determine the volume of the box. Include units.

Owen and Noah are calculating the volume of a rectangular prism with the dimensions of 2 cm $\times$ 2 cm $\times$ 1 cm using different size cubes. Noah uses cubes with side lengths $\frac{1}{3}$ cm and Owen uses cubes with side lengths $\frac{1}{6}$ cm.

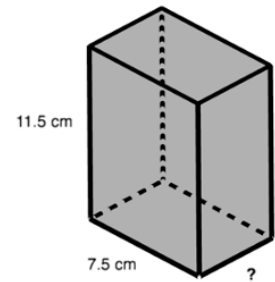
7. Who will need more cubes to fill the larger prism? Justify.
8. What is the volume of the rectangular prism? Include units.
9. If they had used a cube with side lengths $\frac{1}{4}$ cm instead, would they have found the same volume of the rectangular prism?
10. An aquarium in the shape of a rectangular prism is being filled with water. The dimensions of the aquarium are 48.5 inches (in.) by 18.25 in. If the height of the water is 15.75 in., what is the volume of the water in cubic inches?

Grade 6 Accelerated
Unit 8
Lesson 3 Homework

Name _____
Date _____
Period _____

A rectangular prism is shown with measurements in centimeters (cm).

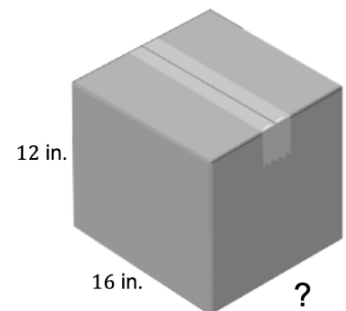
1. What is the area of the front face of this prism? Include units.



2. If the volume of the box is 345 cubic centimeters (cm^3), what is the value of the missing dimension of the box?

Stefani is packing moving boxes. The boxes have a length of 16 inches (in.) and a height of 12 in., but she cannot recall the width of the box.

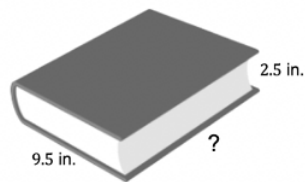
3. Determine the area of the front face of the box. Include units.



4. If the volume of the box is 2,304 cubic inches (in^3), what is the width of the box? Include units.

Jeremiah was trying to find the missing dimension of the journal shown. He made a mistake when solving for the width of the box, but he cannot find it. His work is shown.

Volume = 285 cubic in.



Jeremiah's work	
$\begin{array}{r} 9.5 \\ \times 2.5 \\ \hline 455 \\ + 1900 \\ \hline 2355 \rightarrow 23.55 \text{ in}^2 \end{array}$	$285 \div 23.55 = 12.1$ $w = 12.1 \text{ in.}$

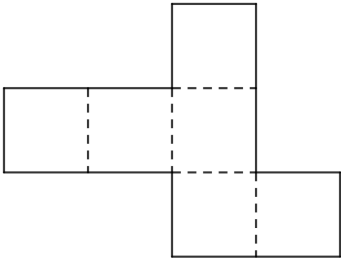
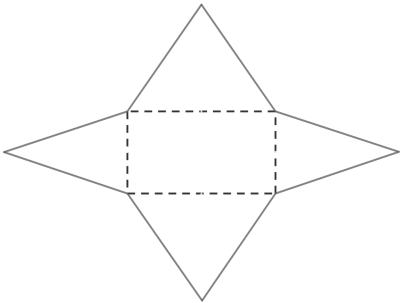
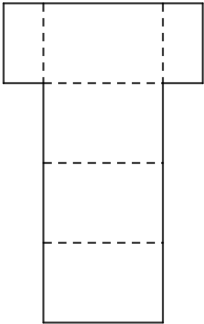
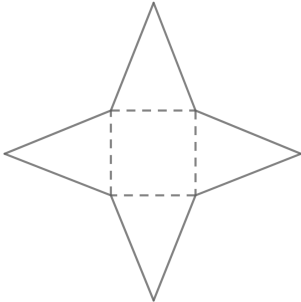
5. Circle the mistake.
6. What should the width of the box be? Include units.
7. Write a note to Jeremiah on how to correct his error(s).
8. A box of crayons has a length of 1.5 inches (in.) and a height of 4 in. The box has a volume of 16.8 cubic inches (in³). What is the width of the box? Include units.
9. Josh bought a sandbox. He knows that the length is 3.5 feet (ft) and the width is 2.5 ft., but he can't remember the appropriate height. The volume needs to be 10.5 cubic feet (ft³). What is the height of the sandbox? Include units.
10. Cara is measuring her suitcase before her trip. The case has a width of 8.75 inches (in.) and a height of 14 in. The volume of the case is 2,756.25 cubic inches (in³). Determine the length of the suitcase. Include units.



Grade 6 Accelerated
Unit 8
Lesson 4 Homework

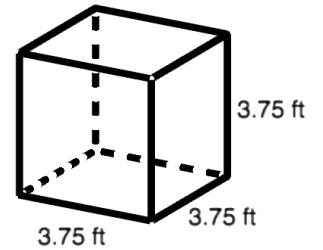
Name _____
Date _____
Period _____

Complete the table to describe the three-dimensional figure formed for questions 1–4.

	Net	Three-Dimensional Figure Formed
1.		
2.		
3.		
4.		

A three-dimensional figure shown for questions 5–6.

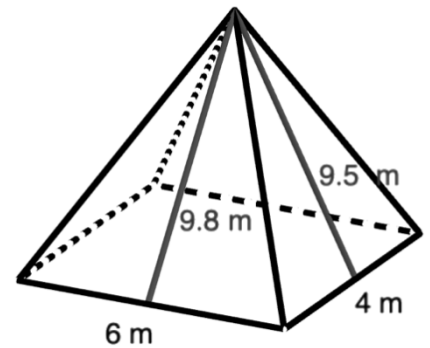
5. Sketch the net or the individual faces of the three-dimensional figure.



6. Label each side of the net with the appropriate measurement in feet (ft).

A three-dimensional figure is shown for questions 7–8.

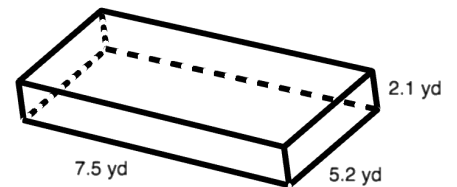
7. Sketch the net or the individual faces of the three-dimensional figure.



8. Label each side of the net with the appropriate measurement in meters (m).

A three-dimensional figure is shown for questions 9–10.

9. Sketch the net or the individual faces of the three-dimensional figure.



10. Label each side of the net with the appropriate measurement in yards (yd).

Grade 6 Accelerated
Unit 8
Lesson 5 Homework

Name _____
Date _____
Period _____

A box of cereal measures 8.8 inches (in.) long, 3.75 in. wide, and 12.75 in. tall.

1. Sketch the net of the cereal box.



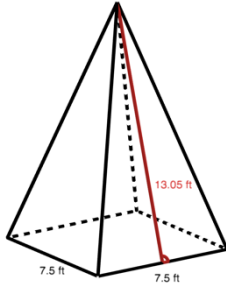
2. Label the dimensions of each face on the net.
3. Find the area of each face. Include units.
4. Add the area of the faces to find the surface area of the cereal box. Include units.

Sharee is designing boxes for the rings she sells. The ring box has measurements 6 centimeters (cm) long, 6 cm wide, and 6 cm high.

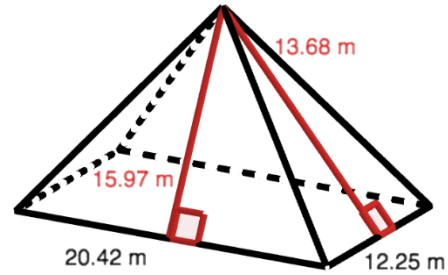
5. Sketch and label the net of the ring box.
6. What is the surface area of the ring box? Include units.

For questions 7–8, find the surface area each pyramid. Include units.

7.



8.



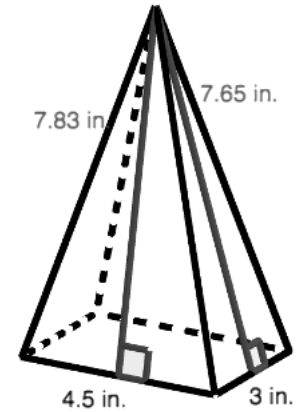
9. Calculate the surface area of a rectangular pyramid that has a square base with a side length of 14.5 centimeters (cm), which is the base of the triangle with a slant height of 14.2 cm.
10. Calculate the surface area of a rectangular pyramid that has a rectangular base with a side length of 12 yards (yd), which is the base of a triangle with a slant height of 11.66 yd, and a side length of 15 yd, which is the base of a triangle with a slant height of 12.5 yd.

Grade 6 Accelerated
Unit 8
Lesson 6 Homework

Name _____
Date _____
Period _____

A rectangular pyramid is shown with measurements in inches (in.).

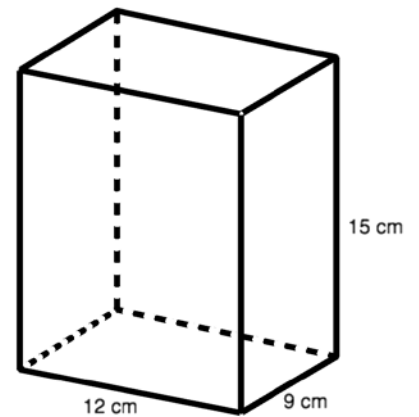
1. Draw a net or the individual faces of the figure. Label the measurements.



2. Find the surface area of the figure.

A rectangular prism is shown with measurements in centimeters (cm).

3. Draw a net or the individual faces of the figure. Label the measurements.



4. Find the surface area of the figure.

5. Find the volume of the figure.

Sanjay is designing a suitcase. He drew a model of his suitcase.

6. If the case has a volume of 7,098 cubic inches (in^3), what is the length of the case?

7. How much material would Sanjay need to make a case with these dimensions, not taking into consideration any overlap of the faces?



8. If the inside of the case has a length of 18 in., a width of 11.5 in., and a height of 24 in., how much smaller is the volume of the inside of the case than the volume of the entire case?

Jocelyn makes and sells bookends that are in the shape of rectangular pyramids.

9. What is the surface area of one bookend?

10. If Jocelyn paints only the triangular faces, what is the surface area of the area she wants to paint?

